Getting Started with Python

Installing Python, pandas, NumPy, and Matplotlib

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This guide walks you, step by step, through installing Python on your computer and setting up three of the most important libraries for data work: **pandas**, **NumPy**, and **Matplotlib**. By the end, you'll have a working Python environment and will have run your first small data script.

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1 Installing Python

Python is the programming language itself. You need to install it before you can install any libraries or run any scripts.

1.1 Step 1: Download the Installer

1. Open your web browser and go to `https://www.python.org/downloads/`
2. The website automatically detects your operating system and shows a **“Download Python 3.x.x”** button. Click it.
3. The installer file will download (e.g., `python-3.12.x-amd64.exe` on Windows, or a `.pkg` file on macOS).

1.2 Step 2: Run the Installer

On Windows:

1. Double-click the downloaded `.exe` file.
2. **Important:** Check the box at the bottom that says *“Add python.exe to PATH”* before clicking Install. This lets you run Python from any terminal window.
3. Click **“Install Now”** and wait for the process to finish.

On macOS:

1. Double-click the downloaded `.pkg` file.
2. Follow the on-screen instructions, agreeing to the license and clicking **Continue/Install**.
3. Enter your Mac password if prompted.

On Linux (Debian/Ubuntu):
Most Linux distributions already include Python. To install or update it, open a terminal and run:

```
sudo apt update
sudo apt install python3 python3-pip
```

Common Pitfall

On Windows, forgetting to check “Add python.exe to PATH” is the single most common setup mistake. If you skip it, typing `python` in a terminal will give you an error like *“python is not recognized”*. You can fix this later by re-running the installer and choosing **Modify**.

1.3 Step 3: Verify the Installation

Open a terminal (**Command Prompt** or **PowerShell** on Windows, **Terminal** on macOS/Linux) and type:

```
python --version
```

You should see output similar to:

```
Python 3.12.4
```

Tip

On macOS and Linux, the command is sometimes `python3` instead of `python`. If `python --version` doesn't work, try `python3 --version`.

Also confirm that `pip`, Python's package installer, is available:

```
pip --version
```

`pip` is installed automatically with Python and is what you'll use in the next section to install pandas, NumPy, and Matplotlib.

2 Installing pandas, NumPy, and Matplotlib

With Python installed, you can now install the three libraries using `pip` directly from your terminal.

2.1 Step 1: (Recommended) Create a Virtual Environment

A virtual environment keeps the packages for one project separate from the rest of your system. It isn't strictly required, but it's good practice.

```
# Create a virtual environment named "venv"
python -m venv venv

# Activate it -- Windows
venv\Scripts\activate

# Activate it -- macOS/Linux
source venv/bin/activate
```

Once activated, your terminal prompt will usually show `(venv)` at the start of the line.

2.2 Step 2: Install the Libraries

With pip, you can install all three libraries in a single command:

```
pip install pandas numpy matplotlib
```

pip will download each package and its dependencies and install them automatically. This may take a minute or two depending on your internet connection.

Tip

If `pip` isn't recognized, try `pip3` instead, or run `python -m pip install pandas numpy matplotlib`, which works even when the `pip` shortcut isn't on your PATH.

2.3 Step 3: Verify the Libraries Installed Correctly

Start the Python interactive shell by typing `python` (or `python3`) in your terminal, then run:

```
import pandas as pd
import numpy as np
import matplotlib
print(pd.__version__)
print(np.__version__)
print(matplotlib.__version__)
```

If no errors appear and you see three version numbers printed, the installation was successful. Type `exit()` to leave the Python shell.

Common Pitfall

If you see `ModuleNotFoundError: No module named 'pandas'`, it usually means either the install failed silently, or you're running a different Python than the one you installed the packages into (common when a virtual environment isn't activated). Re-activate your virtual environment and re-run the `pip install` command.

3 Your First Script

Let's confirm everything works together by writing a tiny script that uses all three libraries: NumPy to generate data, pandas to organize it into a table, and Matplotlib to plot it.

3.1 Step 1: Create the File

Using any text editor (Notepad, VS Code, etc.), create a new file named `first_script.py` with the following content:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Generate some sample data with NumPy
x = np.arange(1, 11)
y = x ** 2

# Organize it into a pandas DataFrame
data = pd.DataFrame({"x": x, "y_squared": y})
print(data)

# Plot it with Matplotlib
plt.plot(data["x"], data["y_squared"], marker="o")
plt.title("y = x^2")
plt.xlabel("x")
plt.ylabel("y")
plt.savefig("first_plot.png")
plt.show()
```

3.2 Step 2: Run the Script

In your terminal, navigate to the folder containing the file and run:

```
python first_script.py
```

You should see a small table printed in the terminal, and a window should pop up showing a simple curved line plot. A file named `first_plot.png` will also be saved in the same folder.

Tip

If the plot window doesn't appear (this can happen in some remote or headless environments), the `plt.savefig(...)` line still saves the image to disk, so you can open `first_plot.png` directly to see the result.

Congratulations! You now have Python, pandas, NumPy, and Matplotlib installed and working together. From here, a great next step is to explore pandas' `read_csv()` function to load your own data.